

```
from google.colab import drive
from google.colab import files
drive.mount('/content/drive',force_remount=True)
```

Mounted at /content/drive

```
train_dir = r'/content/drive/MyDrive/split/train'
test_dir=r'/content/drive/MyDrive/split/test'
```

```
import tensorflow as tf
from tensorflow.keras import initializers
import os
import numpy as np
from tqdm import tqdm
from skimage.io import imread, imshow
from skimage.transform import resize
import matplotlib.pyplot as plt
import random
seed=14
np.random.seed=seed
img_width=224
img_height=224
img_channels=3
```

```
train_img_dir = os.path.join(train_dir,'images/')
train_mask_dir = os.path.join(train_dir,'masks/')
test_img_dir = os.path.join(test_dir,'images/')
test_mask_dir = os.path.join(test_dir,'masks/')
```

```
print(train_img_dir)
test_img_count = len(os.listdir(test_img_dir))
train_img_count=len(os.listdir(train_img_dir))
train_img_ids = os.listdir(train_img_dir)
test_img_ids = os.listdir(test_img_dir)
train_mask_count=len(os.listdir(train_mask_dir))
train_mask_ids=os.listdir(train_mask_dir)
print('Train image count is',train_img_count,'and their ids
are',train_img_ids)
print('Mask image count is',train_mask_count,'and their ids
are',train_mask_ids)
```

```
/content/drive/MyDrive/split/train/images/
Train image count is 2090 and their ids are ['33_64.png',
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test_img_count=len(os.listdir(test_img_dir))
test_img_ids=os.listdir(test_img_dir)
```

```

test_img_count = len(os.listdir(test_img_dir))
test_img_ids = os.listdir(test_img_dir)
print('Test image count is',test_img_count,'and their ids
are',test_img_ids)

```

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Test image count is 233 and their ids are ['26_21.png', '45_22.png',
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'34_32.png', '9_98.png', '32_87.png', '15_57.png', '41_67.png',
'1_89.png', '24_39.png', '41_17.png', '12_3.png', '3_96.png',
'24_32.png', '28_33.png', '33_40.png', '9_75.png', '43_4.png',
'34_72.png', '32_120.png', '26_17.png', '41_65.png', '28_23.png',
'26_109.png', '32_14.png', '23_11.png', '26_58.png', '43_38.png',
'32_109.png', '47_50.png', '21_66.png', '37_41.png', '36_34.png',
'32_64.png', '37_77.png', '18_48.png', '28_56.png', '19_46.png',
'47_10.png', '37_16.png', '16_33.png', '17_45.png', '4_41.png',
'42_76.png', '21_85.png', '16_27.png', '32_84.png', '23_105.png',
'18_58.png', '31_34.png', '23_36.png', '29_58.png', '7_71.png',
'7_5.png', '9_57.png', '24_91.png', '14_57.png', '8_117.png',
'17_31.png', '13_56.png', '18_55.png', '13_78.png', '25_97.png',
'43_61.png', '38_21.png', '20_38.png', '41_63.png', '12_42.png',
'20_3.png', '7_59.png', '19_34.png', '38_23.png', '26_74.png',
'5_106.png', '20_55.png', '31_26.png', '8_110.png', '18_13.png',
'31_115.png', '28_16.png', '9_11.png', '37_54.png', '28_57.png',
'22_72.png', '29_4.png', '22_50.png', '15_58.png', '16_79.png',
'31_113.png', '38_16.png', '13_7.png', '15_78.png', '35_16.png',
'10_48.png', '37_42.png', '31_58.png', '41_62.png', '36_57.png',
'12_105.png', '36_113.png', '11_69.png', '9_86.png', '3_29.png',
'29_40.png', '9_41.png', '3_64.png', '26_69.png', '26_30.png',
'1_29.png', '41_51.png', '37_31.png', '26_68.png', '15_4.png',

```

```

'24_45.png', '34_66.png', '11_77.png', '27_112.png', '9_52.png',
'42_96.png', '3_8.png', '17_119.png', '30_98.png', '41_91.png',
'12_41.png']

X_train =
np.zeros((train_mask_count,img_width,img_height,img_channels),dtype=np
.uint8)
y_train =
np.zeros((len(train_mask_ids),img_width,img_height,1),dtype=np.bool_)

print('Resizing train images and masks')
for n,img_name in
tqdm(enumerate(train_mask_ids),total=len(train_mask_ids)):
    path = train_img_dir + img_name
    img = imread(path)

    # Handle grayscale images by converting to 3 channels if necessary
    if img.ndim == 2:
        img = np.stack([img, img, img], axis=-1)

    # Ensure only img_channels are used
    img = img[:,:,:img_channels]

    img = resize(img,
(img_width,img_height),mode='constant',preserve_range=True)
    X_train[n] = img
    mask = np.zeros((img_width,img_width,1),dtype=np.bool_)
    msk = imread(train_mask_dir+img_name)

    # Ensure mask is 2D if it's 3D (e.g., RGB masks)
    if msk.ndim == 3:
        msk = msk[:, :, 0]

    msk = np.expand_dims(resize(msk,
(img_width,img_height),mode='constant',preserve_range=True),axis=-1)
    mask = np.maximum(mask,msk)
    y_train[n] = mask

```

Resizing train images and masks

100%|██████████| 2090/2090 [22:19<00:00, 1.56it/s]

```

X_test =
np.zeros((test_img_count,img_width,img_height,img_channels),dtype=np.u
int8)
y_test_ground_truth =
np.zeros((len(test_img_ids),img_width,img_height,1),dtype=np.bool_)

print('Resizing test images')
for i,img_name in
tqdm(enumerate(test_img_ids),total=len(test_img_ids)):

```

```

path = test_img_dir + img_name
img = imread(path)

# Handle grayscale images by converting to 3 channels if necessary
if img.ndim == 2:
    img = np.stack([img, img, img], axis=-1)

# Ensure only img_channels are used
img = img[:, :, :img_channels]

img = resize(img,
              (img_width, img_height), mode='constant', preserve_range=True)
X_test[i] = img

```

Resizing test images

```
100%|██████████| 233/233 [02:21<00:00, 1.64it/s]
```

```

print('Resizing test masks')
for n, img_name in
tqdm(enumerate(test_img_ids), total=len(test_img_ids)):
    mask = np.zeros((img_width, img_width, 1), dtype=np.bool_)
    msk = imread(test_mask_dir+img_name)
    msk = np.expand_dims(resize(msk,
                                (img_width, img_height), mode='constant', preserve_range=True), axis=-1)
    mask = np.maximum(mask, msk)
    y_test_ground_truth[n] = mask

```

Resizing test masks

```
100%|██████████| 233/233 [02:17<00:00, 1.70it/s]
```

```
input_shape = (img_height, img_width, img_channels)
```

```

import tensorflow as tf
from tensorflow.keras import layers

```

```

def AttentionGate(x, g, inter_channels):
    """
    x: Encoder feature map (skip connection)
    g: Decoder feature map (gating signal)
    inter_channels: Intermediate channel size
    """

    theta_x = layers.Conv2D(inter_channels, (1,1), padding='same')(x)
    phi_g = layers.Conv2D(inter_channels, (1,1), padding='same')(g)

    add = layers.Add()([theta_x, phi_g])
    relu = layers.Activation('relu')(add)

    psi = layers.Conv2D(1, (1,1), padding='same')(relu)

```

```

sigmoid = layers.Activation('sigmoid')(psi)

attended_x = layers.Multiply()([x, sigmoid])
return attended_x

from tensorflow.keras.applications import VGG19

def VGG19_Encoder(input_tensor):
    base_model = VGG19(weights='imagenet', include_top=False,
input_tensor=input_tensor)

    skips = [
        base_model.get_layer("block1_conv2").output,
        base_model.get_layer("block2_conv2").output,
        base_model.get_layer("block3_conv4").output,
        base_model.get_layer("block4_conv4").output
    ]

    encoder_output = base_model.get_layer("block5_conv4").output

    return skips, encoder_output, base_model

def DecoderBlock(x, skip, filters):
    x = layers.Conv2DTranspose(filters, (2,2), strides=(2,2),
padding='same')(x)

    # Attention Gate applied here
    skip = AttentionGate(skip, x, filters // 2)

    x = layers.Concatenate()([x, skip])

    x = layers.Conv2D(filters, (3,3), padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.Activation('relu')(x)

    x = layers.Conv2D(filters, (3,3), padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.Activation('relu')(x)

    return x

def Attention_VGG19_UNet(input_shape=(224,224,3)):
    inputs = layers.Input(input_shape)

    skips, encoder_output, base_model = VGG19_Encoder(inputs)

    # Freeze early layers
    for layer in base_model.layers[:15]:
        layer.trainable = False

    x = encoder_output

```

```

x = DecoderBlock(x, skips[3], 512)
x = DecoderBlock(x, skips[2], 256)
x = DecoderBlock(x, skips[1], 128)
x = DecoderBlock(x, skips[0], 64)

outputs = layers.Conv2D(1, (1,1), activation='sigmoid')(x)

model = tf.keras.Model(inputs, outputs)
return model

def dice_loss(y_true, y_pred, smooth=1):
    y_true = tf.cast(y_true, tf.float32)
    intersection = tf.reduce_sum(y_true * y_pred)
    return 1 - (2. * intersection + smooth) / (
        tf.reduce_sum(y_true) + tf.reduce_sum(y_pred) + smooth)

def combined_loss(y_true, y_pred):
    return 0.7 * dice_loss(y_true, y_pred) + 0.3 *
tf.keras.losses.binary_crossentropy(y_true, y_pred)

model = Attention_VGG19_UNet()
model.compile(
    optimizer=tf.keras.optimizers.Adam(1e-4),
    loss=combined_loss,
    metrics=['accuracy']
)

model.summary()

```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/vgg19/vgg19_weights_tf_dim_ordering_tf_kernels_notop.h5
80134624/80134624 ————— 4s 0us/step

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
block1_conv1 input_layer[0][0] (Conv2D)	(None, 224, 224, 64)	1,792	

block1_conv2	(None, 224, 224,	36,928	
block1_conv1[0][...	(Conv2D)	64)	
block1_pool	(None, 112, 112,	0	
block1_conv2[0][...	(MaxPooling2D)	64)	
block2_conv1	(None, 112, 112,	73,856	
block1_pool[0][0]	(Conv2D)	128)	
block2_conv2	(None, 112, 112,	147,584	
block2_conv1[0][...	(Conv2D)	128)	
block2_pool	(None, 56, 56,	0	
block2_conv2[0][...	(MaxPooling2D)	128)	
block3_conv1	(None, 56, 56,	295,168	
block2_pool[0][0]	(Conv2D)	256)	
block3_conv2	(None, 56, 56,	590,080	
block3_conv1[0][...	(Conv2D)	256)	
block3_conv3	(None, 56, 56,	590,080	
block3_conv2[0][...	(Conv2D)	256)	

block3_conv4	(None, 56, 56,	590,080	
block3_conv3[0][...]	(Conv2D)	256)	
block3_pool	(None, 28, 28,	0	
block3_conv4[0][...]	(MaxPooling2D)	256)	
block4_conv1	(None, 28, 28,	1,180,160	
block3_pool[0][0]	(Conv2D)	512)	
block4_conv2	(None, 28, 28,	2,359,808	
block4_conv1[0][...]	(Conv2D)	512)	
block4_conv3	(None, 28, 28,	2,359,808	
block4_conv2[0][...]	(Conv2D)	512)	
block4_conv4	(None, 28, 28,	2,359,808	
block4_conv3[0][...]	(Conv2D)	512)	
block4_pool	(None, 14, 14,	0	
block4_conv4[0][...]	(MaxPooling2D)	512)	
block5_conv1	(None, 14, 14,	2,359,808	
block4_pool[0][0]	(Conv2D)	512)	

block5_conv2 block5_conv1[0][...	(None, 14, 14,	2,359,808	
(Conv2D)	512)		
block5_conv3 block5_conv2[0][...	(None, 14, 14,	2,359,808	
(Conv2D)	512)		
block5_conv4 block5_conv3[0][...	(None, 14, 14,	2,359,808	
(Conv2D)	512)		
conv2d_transpose block5_conv4[0][...	(None, 28, 28,	1,049,088	
(Conv2DTranspose)	512)		
conv2d (Conv2D) block4_conv4[0][...	(None, 28, 28,	131,328	
	256)		
conv2d_1 (Conv2D) conv2d_transpose...	(None, 28, 28,	131,328	
	256)		
add (Add)	(None, 28, 28,	0	conv2d[0][0],
	256)		conv2d_1[0]
[0]			
activation	(None, 28, 28,	0	add[0][0]
(Activation)	256)		

conv2d_2 (Conv2D)	(None, 28, 28, 1)	257	activation[0]
activation_1	(None, 28, 28, 1)	0	conv2d_2[0]
(Activation)			
multiply (Multiply)	(None, 28, 28,	0	
block4_conv4[0][...	512)		
activation_1[0][...			
concatenate	(None, 28, 28,	0	
conv2d_transpose...	1024)		
(Concatenate)			multiply[0]
conv2d_3 (Conv2D)	(None, 28, 28,	4,719,104	
concatenate[0][0]	512)		
batch_normalization	(None, 28, 28,	2,048	conv2d_3[0]
(BatchNormalizatio...	512)		
activation_2	(None, 28, 28,	0	
batch_normalizat...	512)		
(Activation)			
conv2d_4 (Conv2D)	(None, 28, 28,	2,359,808	
activation_2[0][...	512)		
batch_normalizatio...	(None, 28, 28,	2,048	conv2d_4[0]
(BatchNormalizatio...	512)		

activation_3 batch_normalizat... (Activation)	(None, 28, 28, 512)	0	
conv2d_transpose_1 activation_3[0][... (Conv2DTranspose)	(None, 56, 56, 256)	524,544	
conv2d_5 (Conv2D) block3_conv4[0][... (Conv2D)	(None, 56, 56, 128)	32,896	
conv2d_6 (Conv2D) conv2d_transpose... (Conv2D)	(None, 56, 56, 128)	32,896	
add_1 (Add) [0], [0]	(None, 56, 56, 128)	0	conv2d_5[0] conv2d_6[0]
activation_4 (Activation)	(None, 56, 56, 128)	0	add_1[0][0]
conv2d_7 (Conv2D) activation_4[0][... (Conv2D)	(None, 56, 56, 1) 129		
activation_5 [0] (Activation)	(None, 56, 56, 1)	0	conv2d_7[0]

multiply_1 block3_conv4[0][...	(None, 56, 56,	0	
(Multiply)	256)		
activation_5[0][...			
concatenate_1 conv2d_transpose...	(None, 56, 56,	0	
(Concatenate)	512)		multiply_1[0]
[0]			
conv2d_8 (Conv2D) concatenate_1[0]...	(None, 56, 56,	1,179,904	
	256)		
batch_normalizatio...	(None, 56, 56,	1,024	conv2d_8[0]
[0]			
(BatchNormalizatio...	256)		
activation_6 batch_normalizat...	(None, 56, 56,	0	
(Activation)	256)		
conv2d_9 (Conv2D) activation_6[0][...	(None, 56, 56,	590,080	
	256)		
batch_normalizatio...	(None, 56, 56,	1,024	conv2d_9[0]
[0]			
(BatchNormalizatio...	256)		
activation_7 batch_normalizat...	(None, 56, 56,	0	
(Activation)	256)		
conv2d_transpose_2	(None, 112, 112,	131,200	

activation_7[0][... (Conv2DTranspose)	128)			
conv2d_10 (Conv2D) block2_conv2[0][...	(None, 112, 112, 64)	8,256		
conv2d_11 (Conv2D) conv2d_transpose...	(None, 112, 112, 64)	8,256		
add_2 (Add) [0], [0]	(None, 112, 112, 64)	0	conv2d_10[0] conv2d_11[0]	
activation_8 (Activation)	(None, 112, 112, 64)	0	add_2[0][0]	
conv2d_12 (Conv2D) activation_8[0][...	(None, 112, 112, 1)	65		
activation_9 [0] (Activation)	(None, 112, 112, 1)	0	conv2d_12[0]	
multiply_2 block2_conv2[0][... (Multiply) activation_9[0][...	(None, 112, 112, 128)	0		
concatenate_2 conv2d_transpose...	(None, 112, 112,	0		

(Concatenate)	256)		multiply_2[0]
[0]			
conv2d_13 (Conv2D)	(None, 112, 112,	295,040	
concatenate_2[0]...	128)		
batch_normalization...	(None, 112, 112,	512	conv2d_13[0]
[0]	(BatchNormalization...	128)	
activation_10	(None, 112, 112,	0	
batch_normalization...	(Activation)	128)	
conv2d_14 (Conv2D)	(None, 112, 112,	147,584	
activation_10[0]...	128)		
batch_normalization...	(None, 112, 112,	512	conv2d_14[0]
[0]	(BatchNormalization...	128)	
activation_11	(None, 112, 112,	0	
batch_normalization...	(Activation)	128)	
conv2d_transpose_3	(None, 224, 224,	32,832	
activation_11[0]...	(Conv2DTranspose)	64)	
conv2d_15 (Conv2D)	(None, 224, 224,	2,080	
block1_conv2[0]...	32)		

conv2d_16 (Conv2D)	(None, 224, 224,	2,080	
conv2d_transpose...	32)		
add_3 (Add)	(None, 224, 224,	0	conv2d_15[0]
[0],	32)		conv2d_16[0]
[0]			
activation_12	(None, 224, 224,	0	add_3[0][0]
(Activation)	32)		
conv2d_17 (Conv2D)	(None, 224, 224,	33	
activation_12[0]...	1)		
activation_13	(None, 224, 224,	0	conv2d_17[0]
[0]			
(Activation)	1)		
multiply_3	(None, 224, 224,	0	
block1_conv2[0][...			
(Multiply)	64)		
activation_13[0]...			
concatenate_3	(None, 224, 224,	0	
conv2d_transpose...			
(Concatenate)	128)		multiply_3[0]
[0]			
conv2d_18 (Conv2D)	(None, 224, 224,	73,792	
concatenate_3[0]...			
	64)		

batch_normalization_14	(None, 224, 224,	256	conv2d_18[0]	
(BatchNormalization)	64)			
activation_14	(None, 224, 224,	0		
batch_normalization_14	(Activation)	64)		
conv2d_19 (Conv2D)	(None, 224, 224,	36,928		
activation_14[0]	64)			
batch_normalization_15	(None, 224, 224,	256	conv2d_19[0]	
(BatchNormalization)	64)			
activation_15	(None, 224, 224,	0		
batch_normalization_15	(Activation)	64)		
conv2d_20 (Conv2D)	(None, 224, 224,	65		
activation_15[0]	1)			

Total params: 31,521,637 (120.25 MB)

Trainable params: 23,292,453 (88.85 MB)

Non-trainable params: 8,229,184 (31.39 MB)

```
import tensorflow as tf
from tensorflow.keras import layers, models
```

```
# -----
# CBAM BLOCK
```

```

# -----
def cbam_block(x, ratio=8):
    channel = x.shape[-1]

    # ----- Channel Attention -----
    avg_pool = layers.GlobalAveragePooling2D()(x)
    max_pool = layers.GlobalMaxPooling2D()(x)

    shared_dense_1 = layers.Dense(channel // ratio, activation='relu')
    shared_dense_2 = layers.Dense(channel)

    avg_out = shared_dense_2(shared_dense_1(avg_pool))
    max_out = shared_dense_2(shared_dense_1(max_pool))

    channel_attention = layers.Activation('sigmoid')(avg_out +
max_out)
    channel_attention = layers.Reshape((1, 1, channel))
(channel_attention)

    x = layers.Multiply()([x, channel_attention])

    # ----- Spatial Attention -----
    avg_pool_spatial = layers.Lambda(lambda z: tf.reduce_mean(z,
axis=-1, keepdims=True))(x)
    max_pool_spatial = layers.Lambda(lambda z: tf.reduce_max(z, axis=-
1, keepdims=True))(x)
    concat = layers.Concatenate(axis=-1)([avg_pool_spatial,
max_pool_spatial])

    spatial_attention = layers.Conv2D(
    1, kernel_size=7, padding='same', activation='sigmoid'
)(concat)

    x = layers.Multiply()([x, spatial_attention])
    return x

# -----
# RESIDUAL CONV BLOCK
# -----
def residual_block(x, filters):
    shortcut = layers.Conv2D(filters, 1, padding='same')(x)
    shortcut = layers.BatchNormalization()(shortcut)

    x = layers.Conv2D(filters, 3, padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.Activation('relu')(x)

    x = layers.Conv2D(filters, 3, padding='same')(x)
    x = layers.BatchNormalization()(x)

```

```

    x = layers.Add()([x, shortcut])
    x = layers.Activation('relu')(x)
    return x

# -----
# RESIDUAL UNET WITH CBAM SKIPS
# -----
def ResidualUNet_CBAM(
    input_shape=(256, 256, 3),
    num_classes=1
):
    inputs = layers.Input(input_shape)

    # ----- Encoder -----
    e1 = residual_block(inputs, 64)
    p1 = layers.MaxPooling2D()(e1)

    e2 = residual_block(p1, 128)
    p2 = layers.MaxPooling2D()(e2)

    e3 = residual_block(p2, 256)
    p3 = layers.MaxPooling2D()(e3)

    e4 = residual_block(p3, 512)
    p4 = layers.MaxPooling2D()(e4)

    # ----- Bottleneck -----
    b = residual_block(p4, 1024)

    # ----- Decoder -----
    d4 = layers.UpSampling2D()(b)
    e4_cbam = cbam_block(e4)
    d4 = layers.Concatenate()([d4, e4_cbam])
    d4 = residual_block(d4, 512)

    d3 = layers.UpSampling2D()(d4)
    e3_cbam = cbam_block(e3)
    d3 = layers.Concatenate()([d3, e3_cbam])
    d3 = residual_block(d3, 256)

    d2 = layers.UpSampling2D()(d3)
    e2_cbam = cbam_block(e2)
    d2 = layers.Concatenate()([d2, e2_cbam])
    d2 = residual_block(d2, 128)

    d1 = layers.UpSampling2D()(d2)
    e1_cbam = cbam_block(e1)
    d1 = layers.Concatenate()([d1, e1_cbam])

```

```

d1 = residual_block(d1, 64)

# ----- Output -----
activation = 'sigmoid' if num_classes == 1 else 'softmax'
outputs = layers.Conv2D(num_classes, 1, activation=activation)(d1)

model = models.Model(inputs, outputs, name="ResidualUNet_CBAM")
return model

# -----
# MODEL INSTANTIATION
# -----
model = ResidualUNet_CBAM(
    input_shape=(256, 256, 3),
    num_classes=1
)

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

model.summary()
Model: "ResidualUNet_CBAM"

```

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 256, 256, 3)	0	-
conv2d_1 (Conv2D) input_layer[0][0]	(None, 256, 256, 64)	1,792	
batch_normalization_1 [0]	(None, 256, 256, 64)	256	conv2d_1[0]

activation	(None, 256, 256,	0		
batch_normalizat... (Activation)	64)			
conv2d_2 (Conv2D)	(None, 256, 256,	36,928	activation[0]	
[0]	64)			
conv2d (Conv2D)	(None, 256, 256,	256		
input_layer[0][0]	64)			
batch_normalizatio... [0]	(None, 256, 256,	256	conv2d_2[0]	
(BatchNormalizatio...	64)			
batch_normalization	(None, 256, 256,	256	conv2d[0][0]	
(BatchNormalizatio...	64)			
add (Add)	(None, 256, 256,	0		
batch_normalizat...	64)			
batch_normalizat...				
activation_1	(None, 256, 256,	0	add[0][0]	
(Activation)	64)			
max_pooling2d	(None, 128, 128,	0		
activation_1[0][...	64)			
(MaxPooling2D)				

conv2d_4 (Conv2D)	(None, 128, 128,	73,856	
max_pooling2d[0]...	128)		
batch_normalization_4 [0]	(None, 128, 128,	512	conv2d_4[0]
(BatchNormalization)	128)		
activation_2	(None, 128, 128,	0	
batch_normalization_4 [0]	128)		
(Activation)			
conv2d_5 (Conv2D)	(None, 128, 128,	147,584	
activation_2[0]...	128)		
conv2d_3 (Conv2D)	(None, 128, 128,	8,320	
max_pooling2d[0]...	128)		
batch_normalization_3 [0]	(None, 128, 128,	512	conv2d_5[0]
(BatchNormalization)	128)		
batch_normalization_3 [0]	(None, 128, 128,	512	conv2d_3[0]
(BatchNormalization)	128)		
add_1 (Add)	(None, 128, 128,	0	
batch_normalization_3 [0]	128)		
batch_normalization_3 [0]			

activation_3 (Activation)	(None, 128, 128, 128)	0	add_1[0][0]
max_pooling2d_1 activation_3[0][...] (MaxPooling2D)	(None, 64, 64, 128)	0	
conv2d_7 (Conv2D) max_pooling2d_1[...] (Conv2D)	(None, 64, 64, 256)	295,168	
batch_normalization_7 conv2d_7[0] (BatchNormalization)	(None, 64, 64, 256)	1,024	conv2d_7[0]
activation_4 batch_normalization_7[0] (Activation)	(None, 64, 64, 256)	0	
conv2d_8 (Conv2D) activation_4[0][...] (Conv2D)	(None, 64, 64, 256)	590,080	
conv2d_6 (Conv2D) max_pooling2d_1[...] (Conv2D)	(None, 64, 64, 256)	33,024	
batch_normalization_6 conv2d_8[0] (BatchNormalization)	(None, 64, 64, 256)	1,024	conv2d_8[0]
batch_normalization_6 conv2d_6[0] (BatchNormalization)	(None, 64, 64, 256)	1,024	conv2d_6[0]

[0]	(BatchNormalization)	256		
add_2 (Add)	(None, 64, 64,	0		
batch_normalizat...	256)			
batch_normalizat...				
activation_5	(None, 64, 64,	0	add_2[0][0]	
(Activation)	256)			
max_pooling2d_2	(None, 32, 32,	0		
activation_5[0][...	(MaxPooling2D)	256)		
conv2d_10 (Conv2D)	(None, 32, 32,	1,180,160		
max_pooling2d_2[...	512)			
batch_normalizatio...	(None, 32, 32,	2,048	conv2d_10[0]	
[0]	(BatchNormalization)	512)		
activation_6	(None, 32, 32,	0		
batch_normalizat...	(Activation)	512)		
conv2d_11 (Conv2D)	(None, 32, 32,	2,359,808		
activation_6[0][...	512)			
conv2d_9 (Conv2D)	(None, 32, 32,	131,584		
max_pooling2d_2[...				

	512)		
batch_normalization_11[0]	(None, 32, 32, 512)	2,048	conv2d_11[0]
batch_normalization_9[0]	(None, 32, 32, 512)	2,048	conv2d_9[0]
add_3 (Add)	(None, 32, 32, 512)	0	
batch_normalization_10			
batch_normalization_8			
activation_7 (Activation)	(None, 32, 32, 512)	0	add_3[0][0]
global_average_pooling2d_3[0]	(None, 512)	0	
global_max_pooling2d_3[0]	(None, 512)	0	
dense (Dense)	(None, 64)	32,832	
global_average_pooling2d_2			
global_max_pooling2d_2			
max_pooling2d_3[0]	(None, 16, 16, 512)	0	

dense_1 (Dense)	(None, 512)	33,280	dense[0][0],
			dense[1][0]
conv2d_13 (Conv2D)	(None, 16, 16,	4,719,616	
max_pooling2d_3[...	1024)		
add_5 (Add)	(None, 512)	0	dense_1[0]
[0],			dense_1[1][0]
batch_normalizatio...	(None, 16, 16,	4,096	conv2d_13[0]
[0]			
(BatchNormalizatio...	1024)		
activation_10	(None, 512)	0	add_5[0][0]
(Activation)			
activation_8	(None, 16, 16,	0	
batch_normalizat...			
(Activation)	1024)		
reshape (Reshape)	(None, 1, 1, 512)	0	
activation_10[0]...			
conv2d_14 (Conv2D)	(None, 16, 16,	9,438,208	
activation_8[0][...	1024)		

conv2d_12 (Conv2D)	(None, 16, 16,	525,312	
max_pooling2d_3[...]	1024)		
multiply (Multiply)	(None, 32, 32,	0	
activation_7[0][...]	512)		reshape[0][0]
batch_normalizatio...	(None, 16, 16,	4,096	conv2d_14[0]
[0]	(BatchNormalizatio...	1024)	
batch_normalizatio...	(None, 16, 16,	4,096	conv2d_12[0]
[0]	(BatchNormalizatio...	1024)	
lambda (Lambda)	(None, 32, 32, 1)	0	multiply[0]
[0]			
lambda_1 (Lambda)	(None, 32, 32, 1)	0	multiply[0]
[0]			
add_4 (Add)	(None, 16, 16,	0	
batch_normalizat...	1024)		
batch_normalizat...			
concatenate	(None, 32, 32, 2)	0	lambda[0][0],
(Concatenate)			lambda_1[0]
[0]			
activation_9	(None, 16, 16,	0	add_4[0][0]
(Activation)	1024)		

conv2d_15 (Conv2D)	(None, 32, 32, 1)	99	
concatenate[0][0]			
global_average_pooling2d[0][0]	(None, 256)	0	
(GlobalAveragePooling2D)			
global_max_pooling2d[0][0]	(None, 256)	0	
(GlobalMaxPooling2D)			
up_sampling2d[0][0]	(None, 32, 32, 1024)	0	
(UpSampling2D)			
multiply_1[0], (Multiply)	(None, 32, 32, 512)	0	multiply[0]
			conv2d_15[0]
dense_2 (Dense)	(None, 32)	8,224	
global_average_pooling2d[0]			
global_max_pooling2d[0]			
concatenate_1[0], up_sampling2d[0]	(None, 32, 32, 1536)	0	
(Concatenate)			multiply_1[0]
dense_3 (Dense)	(None, 256)	8,448	dense_2[0]
			dense_2[1][0]
conv2d_17 (Conv2D)	(None, 32, 32, 1)	7,078,400	
concatenate_1[0]			

	512)		
add_7 (Add)	(None, 256)	0	dense_3[0]
[0],			dense_3[1][0]
batch_normalization_13 (Batch Normalization)	(None, 32, 32, 512)	2,048	conv2d_17[0]
activation_13 (Activation)	(None, 256)	0	add_7[0][0]
activation_11 (Activation)	(None, 32, 32, 512)	0	
batch_normalization_11 (Batch Normalization)			
reshape_1 (Reshape)	(None, 1, 1, 256)	0	
activation_13[0]...			
conv2d_18 (Conv2D)	(None, 32, 32, 512)	2,359,808	
activation_11[0]...			
conv2d_16 (Conv2D)	(None, 32, 32, 512)	786,944	
concatenate_1[0]...			
multiply_2 (Multiply)	(None, 64, 64, 256)	0	
activation_5[0]...			
[0]			reshape_1[0]

batch_normalization_18[0] (BatchNormalization)	(None, 32, 32, 512)	2,048	conv2d_18[0]
batch_normalization_16[0] (BatchNormalization)	(None, 32, 32, 512)	2,048	conv2d_16[0]
lambda_2 (Lambda) [0]	(None, 64, 64, 1)	0	multiply_2[0]
lambda_3 (Lambda) [0]	(None, 64, 64, 1)	0	multiply_2[0]
add_6 (Add) batch_normalization_18 batch_normalization_16	(None, 32, 32, 512)	0	
concatenate_2 [0], (Concatenate) [0]	(None, 64, 64, 2)	0	lambda_2[0] lambda_3[0]
activation_12 (Activation)	(None, 32, 32, 512)	0	add_6[0][0]
conv2d_19 (Conv2D) concatenate_2[0]...	(None, 64, 64, 1)	99	
global_average_pooling_3 activation_3[0]... (GlobalAveragePooling2D)	(None, 128)	0	
global_max_pooling_3	(None, 128)	0	

activation_3[0][...	(GlobalMaxPooling2...			
up_sampling2d_1	(None, 64, 64,	0		
activation_12[0]...	(UpSampling2D)	512)		
multiply_3	(None, 64, 64,	0	multiply_2[0]	
[0],	(Multiply)	256)		
[0]			conv2d_19[0]	
dense_4 (Dense)	(None, 16)	2,064		
global_average_p...				
global_max_pooli...				
concatenate_3	(None, 64, 64,	0		
up_sampling2d_1[...	(Concatenate)	768)		
[0]			multiply_3[0]	
dense_5 (Dense)	(None, 128)	2,176	dense_4[0]	
[0],				
			dense_4[1][0]	
conv2d_21 (Conv2D)	(None, 64, 64,	1,769,728		
concatenate_3[0]...		256)		
add_9 (Add)	(None, 128)	0	dense_5[0]	
[0],				
			dense_5[1][0]	
batch_normalizatio...	(None, 64, 64,	1,024	conv2d_21[0]	
[0]				

(BatchNormalizatio...	256)		
activation_16	(None, 128)	0	add_9[0][0]
(Activation)			
activation_14	(None, 64, 64,	0	
batch_normalizat...	(Activation)	256)	
reshape_2 (Reshape)	(None, 1, 1, 128)	0	
activation_16[0]...			
conv2d_22 (Conv2D)	(None, 64, 64,	590,080	
activation_14[0]...	256)		
conv2d_20 (Conv2D)	(None, 64, 64,	196,864	
concatenate_3[0]...	256)		
multiply_4	(None, 128, 128,	0	
activation_3[0][...	(Multiply)	128)	reshape_2[0]
[0]			
batch_normalizatio...	(None, 64, 64,	1,024	conv2d_22[0]
[0]	(BatchNormalizatio...	256)	
batch_normalizatio...	(None, 64, 64,	1,024	conv2d_20[0]
[0]	(BatchNormalizatio...	256)	

lambda_4 (Lambda)	(None, 128, 128,	0	multiply_4[0]
[0]	1)		
lambda_5 (Lambda)	(None, 128, 128,	0	multiply_4[0]
[0]	1)		
add_8 (Add)	(None, 64, 64,	0	
batch_normalizat...	256)		
batch_normalizat...			
concatenate_4	(None, 128, 128,	0	lambda_4[0]
[0],			
(Concatenate)	2)		lambda_5[0]
[0]			
activation_15	(None, 64, 64,	0	add_8[0][0]
(Activation)	256)		
conv2d_23 (Conv2D)	(None, 128, 128,	99	
concatenate_4[0]...	1)		
global_average_poo...	(None, 64)	0	
activation_1[0][...			
(GlobalAveragePool...			
global_max_pooling...	(None, 64)	0	
activation_1[0][...			
(GlobalMaxPooling2...			

up_sampling2d_2 activation_15[0]...	(None, 128, 128,	0	
(UpSampling2D)	256)		
multiply_5 [0],	(None, 128, 128,	0	multiply_4[0]
(Multiply)	128)		conv2d_23[0]
[0]			
dense_6 (Dense)	(None, 8)	520	
global_average_p...			
global_max_pooli...			
concatenate_5 up_sampling2d_2[...	(None, 128, 128,	0	
(Concatenate)	384)		multiply_5[0]
[0]			
dense_7 (Dense)	(None, 64)	576	dense_6[0]
[0],			dense_6[1][0]
conv2d_25 (Conv2D)	(None, 128, 128,	442,496	
concatenate_5[0]...	128)		
add_11 (Add)	(None, 64)	0	dense_7[0]
[0],			dense_7[1][0]
batch_normalizatio...	(None, 128, 128,	512	conv2d_25[0]
[0]			
(BatchNormalizatio...	128)		
activation_19	(None, 64)	0	add_11[0][0]

(Activation)			
activation_17 batch_normalizat...	(None, 128, 128,	0	
(Activation)	128)		
reshape_3 (Reshape) activation_19[0]...	(None, 1, 1, 64)	0	
conv2d_26 (Conv2D) activation_17[0]...	(None, 128, 128,	147,584	
	128)		
conv2d_24 (Conv2D) concatenate_5[0]...	(None, 128, 128,	49,280	
	128)		
multiply_6 activation_1[0][... (Multiply)	(None, 256, 256,	0	
[0]	64)		reshape_3[0]
batch_normalizatio... [0]	(None, 128, 128,	512	conv2d_26[0]
(BatchNormalizatio...	128)		
batch_normalizatio... [0]	(None, 128, 128,	512	conv2d_24[0]
(BatchNormalizatio...	128)		
lambda_6 (Lambda) [0]	(None, 256, 256,	0	multiply_6[0]
	1)		

lambda_7 (Lambda)	(None, 256, 256,	0	multiply_6[0]
[0]	1)		
add_10 (Add)	(None, 128, 128,	0	
batch_normalizat...	128)		
batch_normalizat...			
concatenate_6	(None, 256, 256,	0	lambda_6[0]
[0],			
(Concatenate)	2)		lambda_7[0]
[0]			
activation_18	(None, 128, 128,	0	add_10[0][0]
(Activation)	128)		
conv2d_27 (Conv2D)	(None, 256, 256,	99	
concatenate_6[0]...	1)		
up_sampling2d_3	(None, 256, 256,	0	
activation_18[0]...	128)		
(UpSampling2D)			
multiply_7	(None, 256, 256,	0	multiply_6[0]
[0],			
(Multiply)	64)		conv2d_27[0]
[0]			
concatenate_7	(None, 256, 256,	0	
up_sampling2d_3[...	192)		multiply_7[0]
(Concatenate)			
[0]			

conv2d_29 (Conv2D)	(None, 256, 256,	110,656	
concatenate_7[0]...	64)		
batch_normalizatio...	(None, 256, 256,	256	conv2d_29[0]
[0]	(BatchNormalizatio...	64)	
activation_20	(None, 256, 256,	0	
batch_normalizat...	(Activation)	64)	
conv2d_30 (Conv2D)	(None, 256, 256,	36,928	
activation_20[0]...	64)		
conv2d_28 (Conv2D)	(None, 256, 256,	12,352	
concatenate_7[0]...	64)		
batch_normalizatio...	(None, 256, 256,	256	conv2d_30[0]
[0]	(BatchNormalizatio...	64)	
batch_normalizatio...	(None, 256, 256,	256	conv2d_28[0]
[0]	(BatchNormalizatio...	64)	
add_12 (Add)	(None, 256, 256,	0	
batch_normalizat...	64)		
batch_normalizat...			

activation_21	(None, 256, 256,	0	add_12[0][0]
(Activation)	64)		
conv2d_31 (Conv2D)	(None, 256, 256,	65	
activation_21[0]...	1)		

Total params: 33,246,725 (126.83 MB)

Trainable params: 33,229,061 (126.76 MB)

Non-trainable params: 17,664 (69.00 KB)

```
X_train = X_train.astype("float32")
```

```
X_test = X_test.astype("float32")
```

```
# # Normalize if needed
```

```
if X_train.max() > 1.0:
```

```
    X_train /= 255.0
```

```
    X_test /= 255.0
```

```
if y_train.ndim == 3:
```

```
    y_train = np.expand_dims(y_train, axis=-1)
```

```
if y_test_ground_truth.ndim == 3:
```

```
    y_test_ground_truth = np.expand_dims(y_test_ground_truth, axis=-1)
```

```
y_train = y_train.astype("float32")
```

```
y_test_ground_truth = y_test_ground_truth.astype("float32")
```

```
callbacks = [
```

```
    tf.keras.callbacks.ModelCheckpoint(
```

```
        "NeuroHybridNet_best.h5",
```

```
        monitor="val_loss",
```

```
        save_best_only=True,
```

```
        verbose=1
```

```
    ),
```

```
    tf.keras.callbacks.ReduceLROnPlateau(
```

```
        monitor="val_loss",
```

```
        factor=0.2,
```

```
        patience=5,
```

```

        verbose=1
    ),
    tf.keras.callbacks.EarlyStopping(
        monitor="val_loss",
        patience=10,
        restore_best_weights=True,
        verbose=1
    )
]
history = model.fit(
    X_train,
    y_train,
    validation_split=0.2,
    batch_size=4,
    epochs=100,
    callbacks=callbacks,
    shuffle=True
)

```

Epoch 1/100

417/418 ————— 0s 26ms/step - accuracy: 0.8751 - loss: 0.7488

Epoch 1: val_loss improved from inf to 0.87793, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

418/418 ————— 46s 44ms/step - accuracy: 0.8753 - loss: 0.7482 - val_accuracy: 0.7088 - val_loss: 0.8779 - learning_rate: 1.0000e-04

Epoch 2/100

417/418 ————— 0s 26ms/step - accuracy: 0.9812 - loss: 0.3442

Epoch 2: val_loss improved from 0.87793 to 0.25396, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

418/418 ————— 13s 31ms/step - accuracy: 0.9812 - loss: 0.3440 - val_accuracy: 0.9814 - val_loss: 0.2540 - learning_rate: 1.0000e-04

Epoch 3/100
417/418 ————— 0s 26ms/step - accuracy: 0.9868 - loss: 0.2010

Epoch 3: val_loss improved from 0.25396 to 0.23060, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

418/418 ————— 13s 31ms/step - accuracy: 0.9868 - loss: 0.2010 - val_accuracy: 0.9788 - val_loss: 0.2306 - learning_rate: 1.0000e-04

Epoch 4/100
417/418 ————— 0s 26ms/step - accuracy: 0.9883 - loss: 0.1580

Epoch 4: val_loss improved from 0.23060 to 0.16937, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

418/418 ————— 13s 30ms/step - accuracy: 0.9883 - loss: 0.1580 - val_accuracy: 0.9865 - val_loss: 0.1694 - learning_rate: 1.0000e-04

Epoch 5/100
417/418 ————— 0s 26ms/step - accuracy: 0.9892 - loss: 0.1386

Epoch 5: val_loss improved from 0.16937 to 0.16560, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

418/418 ————— 13s 31ms/step - accuracy: 0.9892 - loss: 0.1386 - val_accuracy: 0.9872 - val_loss: 0.1656 - learning_rate: 1.0000e-04

Epoch 6/100
417/418 ————— 0s 26ms/step - accuracy: 0.9900 - loss: 0.1248

Epoch 6: val_loss did not improve from 0.16560

418/418 ————— 12s 29ms/step - accuracy: 0.9900 - loss: 0.1248 - val_accuracy: 0.9856 - val_loss: 0.1659 - learning_rate:


```
1.0000e-04
Epoch 7/100
417/418 _____ 0s 26ms/step - accuracy: 0.9909 - loss:
0.1120
```

Epoch 7: val_loss improved from 0.16560 to 0.15872, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

```
418/418 _____ 13s 31ms/step - accuracy: 0.9909 - loss:
0.1120 - val_accuracy: 0.9870 - val_loss: 0.1587 - learning_rate:
1.0000e-04
```

Epoch 8/100

```
417/418 _____ 0s 26ms/step - accuracy: 0.9914 - loss:
0.1045
```

Epoch 8: val_loss did not improve from 0.15872

```
418/418 _____ 12s 29ms/step - accuracy: 0.9914 - loss:
0.1045 - val_accuracy: 0.9863 - val_loss: 0.1607 - learning_rate:
1.0000e-04
```

Epoch 9/100

```
417/418 _____ 0s 26ms/step - accuracy: 0.9918 - loss:
0.0991
```

Epoch 9: val_loss improved from 0.15872 to 0.15650, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

```
418/418 _____ 13s 30ms/step - accuracy: 0.9918 - loss:
0.0991 - val_accuracy: 0.9870 - val_loss: 0.1565 - learning_rate:
1.0000e-04
```

Epoch 10/100

```
417/418 _____ 0s 26ms/step - accuracy: 0.9919 - loss:
0.0965
```

Epoch 10: val_loss did not improve from 0.15650

```
418/418 _____ 12s 29ms/step - accuracy: 0.9919 - loss:
0.0965 - val_accuracy: 0.9870 - val_loss: 0.1633 - learning_rate:
1.0000e-04
```

Epoch 11/100

```
417/418 _____ 0s 27ms/step - accuracy: 0.9927 - loss:
0.0877
```

Epoch 11: val_loss improved from 0.15650 to 0.15358, saving model to NeuroHybridNet_best.h5

WARNING:absl:You are saving your model as an HDF5 file via
`model.save()` or `keras.saving.save\_model(model)`. This file format
is considered legacy. We recommend using instead the native Keras
format, e.g. `model.save('my\_model.keras')` or
`keras.saving.save\_model(model, 'my\_model.keras')`.

418/418 _____ 13s 31ms/step - accuracy: 0.9927 - loss:
0.0878 - val_accuracy: 0.9873 - val_loss: 0.1536 - learning_rate:
1.0000e-04

Epoch 12/100

417/418 _____ 0s 26ms/step - accuracy: 0.9927 - loss:
0.0861

Epoch 12: val_loss did not improve from 0.15358

418/418 _____ 12s 29ms/step - accuracy: 0.9927 - loss:
0.0861 - val_accuracy: 0.9872 - val_loss: 0.1576 - learning_rate:
1.0000e-04

Epoch 13/100

417/418 _____ 0s 26ms/step - accuracy: 0.9930 - loss:
0.0825

Epoch 13: val_loss did not improve from 0.15358

418/418 _____ 12s 29ms/step - accuracy: 0.9930 - loss:
0.0825 - val_accuracy: 0.9872 - val_loss: 0.1565 - learning_rate:
1.0000e-04

Epoch 14/100

417/418 _____ 0s 27ms/step - accuracy: 0.9935 - loss:
0.0775

Epoch 14: val_loss did not improve from 0.15358

418/418 _____ 12s 29ms/step - accuracy: 0.9935 - loss:
0.0775 - val_accuracy: 0.9866 - val_loss: 0.1611 - learning_rate:
1.0000e-04

Epoch 15/100

417/418 _____ 0s 26ms/step - accuracy: 0.9937 - loss:
0.0747

Epoch 15: val_loss did not improve from 0.15358

418/418 _____ 12s 29ms/step - accuracy: 0.9937 - loss:
0.0747 - val_accuracy: 0.9868 - val_loss: 0.1582 - learning_rate:
1.0000e-04

Epoch 16/100

417/418 _____ 0s 26ms/step - accuracy: 0.9941 - loss:
0.0700

Epoch 16: val_loss did not improve from 0.15358

Epoch 16: ReduceLROnPlateau reducing learning rate to

1.9999999494757503e-05.

418/418 _____ 12s 29ms/step - accuracy: 0.9941 - loss:
0.0700 - val_accuracy: 0.9870 - val_loss: 0.1604 - learning_rate:
1.0000e-04

Epoch 17/100

417/418 _____ 0s 26ms/step - accuracy: 0.9945 - loss:
0.0650

Epoch 17: val_loss did not improve from 0.15358
418/418 ————— 12s 29ms/step - accuracy: 0.9945 - loss: 0.0650 - val_accuracy: 0.9874 - val_loss: 0.1579 - learning_rate: 2.0000e-05

Epoch 18/100

417/418 ————— 0s 26ms/step - accuracy: 0.9952 - loss: 0.0571

Epoch 18: val_loss did not improve from 0.15358

418/418 ————— 12s 29ms/step - accuracy: 0.9952 - loss: 0.0571 - val_accuracy: 0.9873 - val_loss: 0.1596 - learning_rate: 2.0000e-05

Epoch 19/100

417/418 ————— 0s 26ms/step - accuracy: 0.9955 - loss: 0.0530

Epoch 19: val_loss did not improve from 0.15358

418/418 ————— 12s 29ms/step - accuracy: 0.9955 - loss: 0.0530 - val_accuracy: 0.9872 - val_loss: 0.1591 - learning_rate: 2.0000e-05

Epoch 20/100

417/418 ————— 0s 26ms/step - accuracy: 0.9957 - loss: 0.0508

Epoch 20: val_loss did not improve from 0.15358

418/418 ————— 12s 29ms/step - accuracy: 0.9957 - loss: 0.0508 - val_accuracy: 0.9872 - val_loss: 0.1611 - learning_rate: 2.0000e-05

Epoch 21/100

417/418 ————— 0s 26ms/step - accuracy: 0.9959 - loss: 0.0490

Epoch 21: val_loss did not improve from 0.15358

Epoch 21: ReduceLROnPlateau reducing learning rate to 3.999999898951501e-06.

418/418 ————— 12s 29ms/step - accuracy: 0.9959 - loss: 0.0490 - val_accuracy: 0.9873 - val_loss: 0.1611 - learning_rate: 2.0000e-05

Epoch 21: early stopping

Restoring model weights from the end of the best epoch: 11.

```
Y_pred = model.predict(X_test, batch_size=4)
```

```
Y_pred_bin = (Y_pred > 0.3).astype("uint8")
```

59/59 ————— 1s 10ms/step

Flatten arrays

```
y_true = y_test_ground_truth.flatten().astype(np.uint8)
```

```
y_pred = Y_pred_bin.flatten().astype(np.uint8)
```

Confusion matrix components

```
TP = np.sum((y_true == 1) & (y_pred == 1))
```

```
TN = np.sum((y_true == 0) & (y_pred == 0))
```

```

FP = np.sum((y_true == 0) & (y_pred == 1))
FN = np.sum((y_true == 1) & (y_pred == 0))

# Metrics
precision = TP / (TP + FP + 1e-8)
recall    = TP / (TP + FN + 1e-8)
f1_score  = 2 * precision * recall / (precision + recall + 1e-8)
iou       = TP / (TP + FP + FN + 1e-8)
accuracy  = (TP + TN) / (TP + TN + FP + FN)

# Cohen's Kappa
total = TP + TN + FP + FN
po = accuracy
pe = (((TP + FP) * (TP + FN)) + ((FN + TN) * (FP + TN))) / (total ** 2)
kappa = (po - pe) / (1 - pe + 1e-8)

# Print metrics
print("==== PIXEL-LEVEL EVALUATION METRICS =====")
print(f"Precision      : {precision:.4f}")
print(f"Recall         : {recall:.4f}")
print(f"F1-score       : {f1_score:.4f}")
print(f"IoU           : {iou:.4f}")
print(f"Overall Accuracy: {accuracy:.4f}")
print(f"Kappa (κ)      : {kappa:.4f}")
print("=====")

# Confusion Matrix
confusion_matrix = np.array([[TN, FP],
                             [FN, TP]])

print("\n==== CONFUSION MATRIX =====")
print(confusion_matrix)

==== PIXEL-LEVEL EVALUATION METRICS =====
Precision      : 0.8049
Recall         : 0.8064
F1-score       : 0.8056
IoU           : 0.6745
Overall Accuracy: 0.9863
Kappa (κ)      : 0.7985
=====

==== CONFUSION MATRIX =====
[[11199085   80442]
 [  79665   331816]]

import matplotlib.pyplot as plt
import numpy as np

```

```

# Class labels
classes = ["Background", "Nerve"]

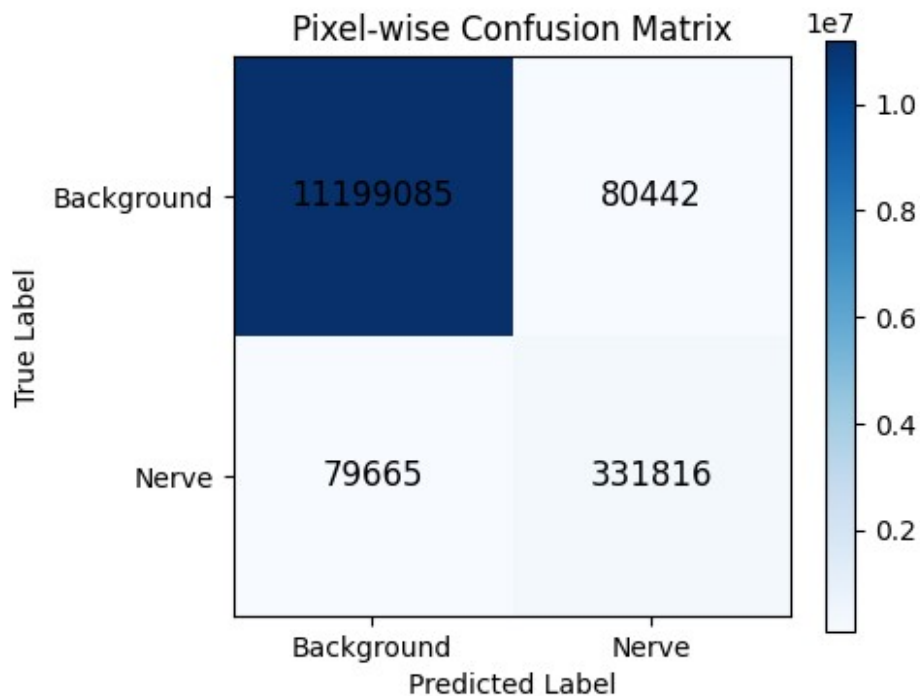
plt.figure(figsize=(5, 4))
plt.imshow(confusion_matrix, cmap="Blues")
plt.title("Pixel-wise Confusion Matrix")
plt.colorbar()

# Axis ticks and labels
plt.xticks([0, 1], classes)
plt.yticks([0, 1], classes)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")

# Annotate each cell with count
for i in range(2):
    for j in range(2):
        plt.text(
            j, i,
            confusion_matrix[i, j],
            ha="center", va="center",
            color="black", fontsize=12
        )

plt.tight_layout()
plt.show()

```



```

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.patches import Rectangle

fig, ax = plt.subplots(figsize=(12, 8))

# Define colors for each cell
cell_colors = [
    ["#4F81BD", "#C0504D"], # TN (Blue), FP (Red)
    ["#F79646", "#9BBB59"] # FN (Orange), TP (Green)
]

# Draw colored cells
for i in range(2):
    for j in range(2):
        ax.add_patch(
            Rectangle(
                (j - 0.5, i - 0.5),
                1, 1,
                facecolor=cell_colors[i][j],
                edgecolor="black"
            )
        )
        ax.text(
            j, i,
            f"{confusion_matrix[i, j]}",
            ha="center", va="center",
            fontsize=13, fontweight="bold", color="black"
        )

# Set ticks and labels
classes = ["Background", "Nerve"]
ax.set_xticks([0, 1])
ax.set_yticks([0, 1])
ax.set_xticklabels(classes)
ax.set_yticklabels(classes)

ax.set_xlabel("Predicted Label")
ax.set_ylabel("True Label")
ax.set_title("Pixel-wise Confusion Matrix")

# Add TP / FP / FN / TN labels
labels = [["TN", "FP"], ["FN", "TP"]]
for i in range(2):
    for j in range(2):
        ax.text(
            j, i + 0.25,
            labels[i][j],
            ha="center", va="center",
            fontsize=10
        )

```

```

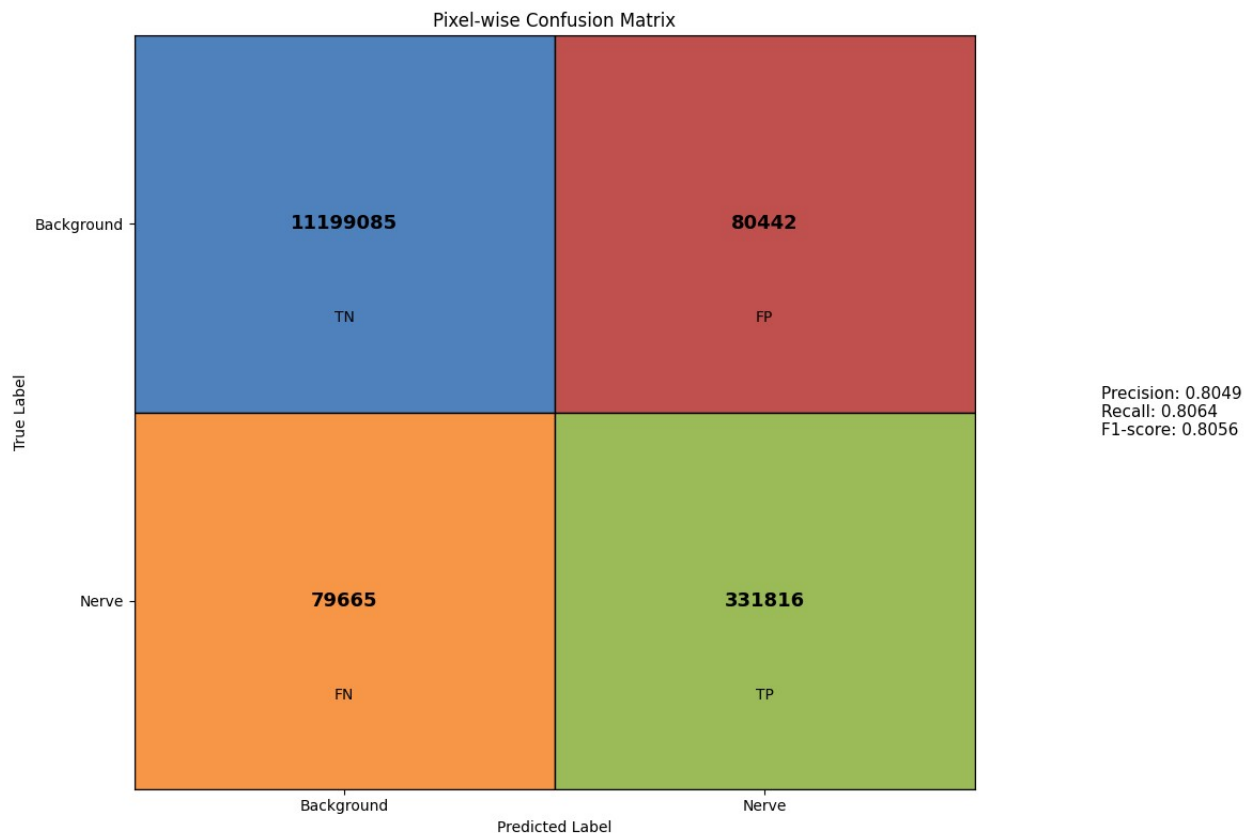
    )

# Add precision & recall text
ax.text(
    1.15, 0.5,
    f"Precision: {precision:.4f}\nRecall: {recall:.4f}\nF1-score: {f1_score:.4f}",
    transform=ax.transAxes,
    fontsize=11,
    verticalalignment="center"
)

# Fix axis limits and grid
ax.set_xlim(-0.5, 1.5)
ax.set_ylim(1.5, -0.5)
ax.grid(False)

plt.tight_layout()
plt.show()

```



```

def overlay_tp_fp_fn(image, y_true, y_pred, alpha=0.6):
    """
    image: (H, W, 3)
    y_true, y_pred: (H, W) binary
    """

```

```

"""
if image.max() > 1:
    image = image / 255.0

y_true = y_true.squeeze().astype(np.uint8)
y_pred = y_pred.squeeze().astype(np.uint8)

overlay = image.copy()

TP = (y_true == 1) & (y_pred == 1)
FP = (y_true == 0) & (y_pred == 1)
FN = (y_true == 1) & (y_pred == 0)

overlay[TP] = (1 - alpha) * overlay[TP] + alpha * np.array([0, 1,
0]) # Green
overlay[FP] = (1 - alpha) * overlay[FP] + alpha * np.array([1, 0,
0]) # Red
overlay[FN] = (1 - alpha) * overlay[FN] + alpha * np.array([0, 0,
1]) # Blue

return overlay

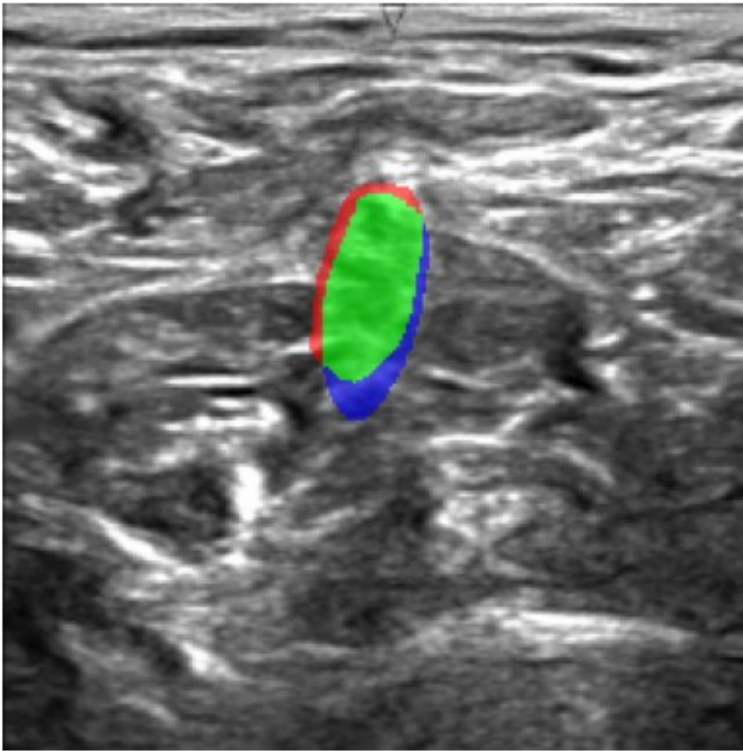
idx = indices[0] # any test index

overlay_img = overlay_tp_fp_fn(
    X_test[idx],
    y_test_ground_truth[idx],
    Y_pred_bin[idx]
)

plt.figure(figsize=(5,5))
plt.imshow(overlay_img)
plt.title("TP (Green) | FP (Red) | FN (Blue)")
plt.axis("off")
plt.show()

```


TP (Green) | FP (Red) | FN (Blue)



```
from sklearn.metrics import precision_recall_curve, auc
import matplotlib.pyplot as plt

y_scores = Y_pred.flatten()          # probabilities
y_true_flat = y_test_ground_truth.flatten()

precision_curve, recall_curve, _ = precision_recall_curve(y_true_flat,
y_scores)
pr_auc = auc(recall_curve, precision_curve)

plt.figure(figsize=(5,4))

# Precision-Recall curve (Blue)
plt.plot(
    recall_curve,
    precision_curve,
    color="blue",
    linewidth=2,
    label=f"PR Curve (AUC = {pr_auc:.4f})"
)

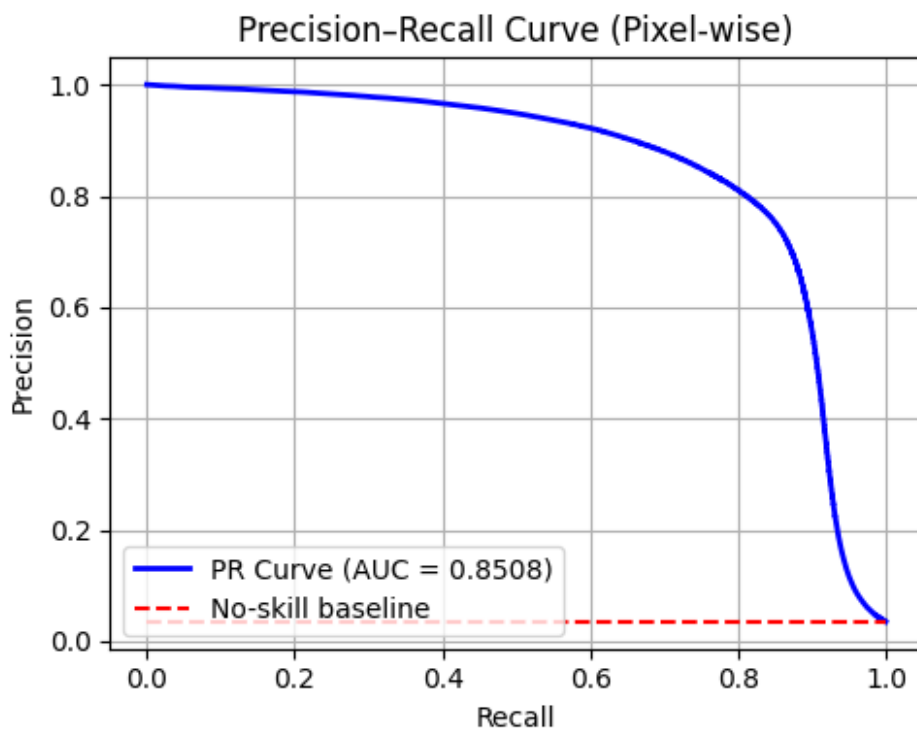
# No-skill baseline (Red)
baseline = y_true_flat.mean()
plt.hlines(
    baseline, 0, 1,
```

```

        colors="red",
        linestyle="--",
        label="No-skill baseline"
    )

plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision-Recall Curve (Pixel-wise)")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```



```

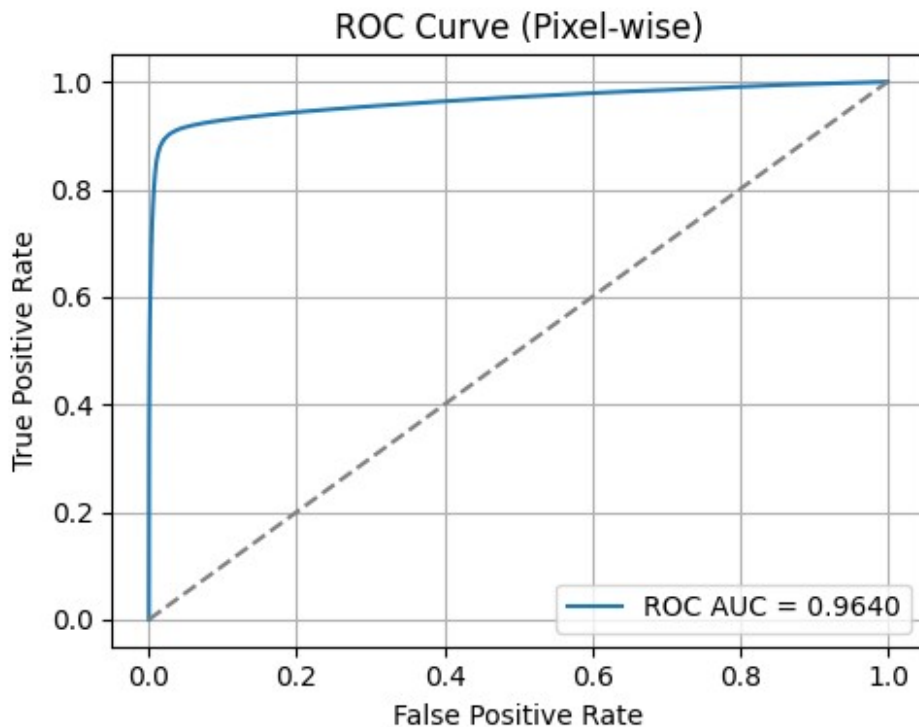
from sklearn.metrics import roc_curve, auc

fpr, tpr, _ = roc_curve(y_true_flat, y_scores)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(5,4))
plt.plot(fpr, tpr, label=f"ROC AUC = {roc_auc:.4f}")
plt.plot([0,1], [0,1], linestyle="--", color="gray")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve (Pixel-wise)")
plt.legend()
plt.grid(True)

```

```
plt.tight_layout()
plt.show()
```



```
import matplotlib.pyplot as plt
import numpy as np

num_images = 40
indices = np.random.choice(len(X_test), num_images, replace=False)

plt.figure(figsize=(12, num_images * 3))

for i, idx in enumerate(indices):
    # Input Image
    plt.subplot(num_images, 3, i*3 + 1)
    plt.imshow(X_test[idx])
    plt.title(f"Input Image {i+1}")
    plt.axis("off")

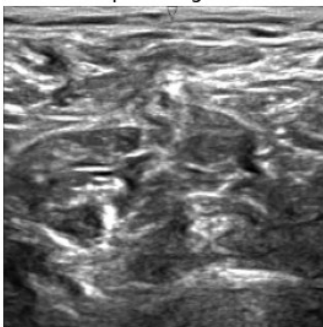
    # Ground Truth
    plt.subplot(num_images, 3, i*3 + 2)
    plt.imshow(y_test_ground_truth[idx].squeeze(), cmap="gray")
    plt.title("Ground Truth")
    plt.axis("off")

    # Prediction
    plt.subplot(num_images, 3, i*3 + 3)
```

```
plt.imshow(Y_pred_bin[idx].squeeze(), cmap="gray")
plt.title("Prediction")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Input Image 1



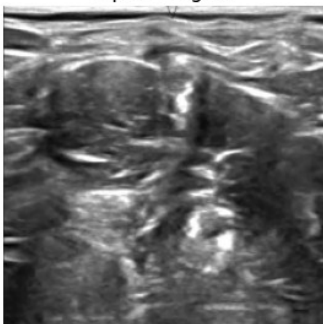
Ground Truth



Prediction



Input Image 2



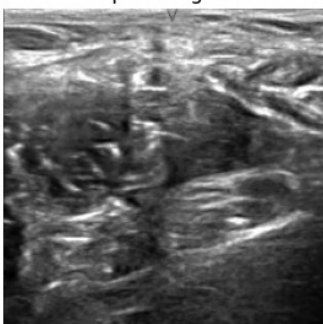
Ground Truth



Prediction



Input Image 3



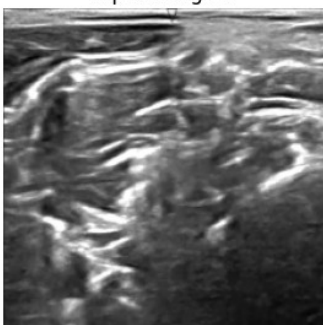
Ground Truth



Prediction



Input Image 4



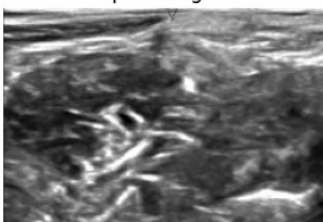
Ground Truth



Prediction



Input Image 5



Ground Truth



Prediction



```

import matplotlib.pyplot as plt

# Create figure
plt.figure(figsize=(18, 5))

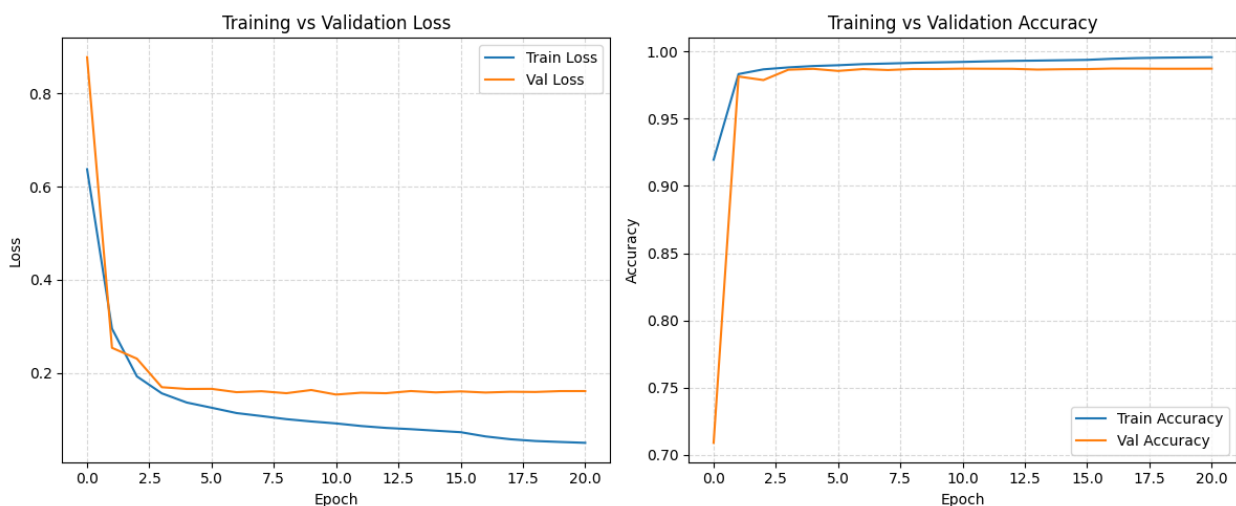
# ---- LOSS ----
plt.subplot(1, 3, 1)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training vs Validation Loss')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)

# ---- ACCURACY ----
plt.subplot(1, 3, 2)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Val Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.title('Training vs Validation Accuracy')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)

# ---- IoU ----

plt.tight_layout()
plt.show()

```



```

import numpy as np

def overlay_mask(image, mask, alpha=0.5):
    """

```

```

image: (H, W, 3) RGB image, range [0,1] or [0,255]
mask:  (H, W) or (H, W, 1), binary (0/1)
alpha: transparency factor
"""

# Ensure image is float in [0,1]
if image.max() > 1:
    image = image / 255.0

# Squeeze mask and binarize
mask = np.squeeze(mask)
mask = (mask > 0.5).astype(np.float32)

# Create red mask
red_mask = np.zeros_like(image)
red_mask[..., 0] = 1.0 # red channel

# Blend image and mask
overlay = image * (1 - alpha * mask[..., None]) + \
    red_mask * (alpha * mask[..., None])

return np.clip(overlay, 0, 1)

import matplotlib.pyplot as plt

num_images = len(indices)

plt.figure(figsize=(12, num_images * 3))

for i, idx in enumerate(indices):
    overlay_img = overlay_mask(X_test[idx], Y_pred_bin[idx])

    # Input image
    plt.subplot(num_images, 3, i * 3 + 1)
    plt.imshow(X_test[idx])
    plt.title("Input Image")
    plt.axis("off")

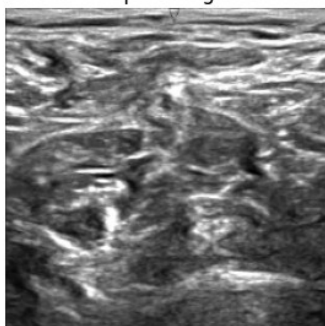
    # Ground truth mask
    plt.subplot(num_images, 3, i * 3 + 2)
    plt.imshow(y_test_ground_truth[idx].squeeze(), cmap="gray")
    plt.title("Ground Truth")
    plt.axis("off")

    # Prediction overlay
    plt.subplot(num_images, 3, i * 3 + 3)
    plt.imshow(overlay_img)
    plt.title("Prediction Overlay")
    plt.axis("off")

```

```
plt.tight_layout()  
plt.show()
```

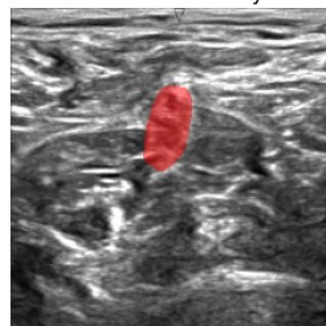

Input Image



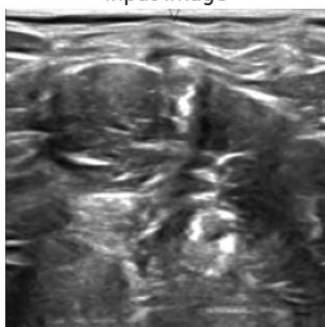
Ground Truth



Prediction Overlay



Input Image



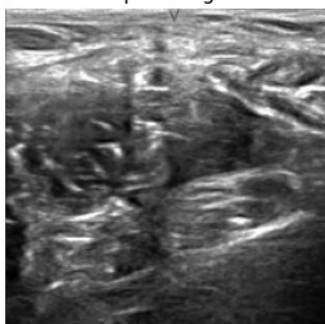
Ground Truth



Prediction Overlay



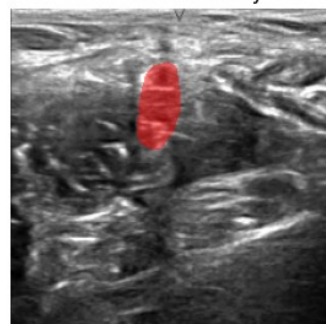
Input Image



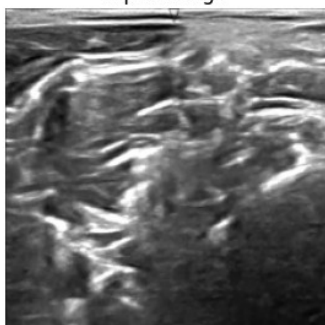
Ground Truth



Prediction Overlay



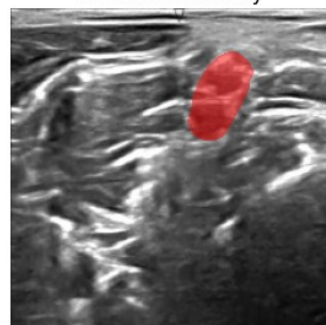
Input Image



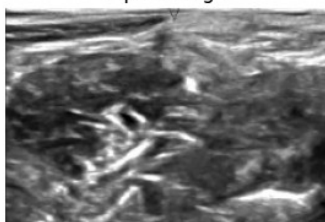
Ground Truth



Prediction Overlay



Input Image



Ground Truth



Prediction Overlay

